Experiment 3: Average Sale Price of Houses with More Than Four Bedrooms

Aim

To find the average sale price of houses with more than four bedrooms using NumPy arrays and conditional filtering.

Algorithm

1. Start the program.
2. Import the NumPy library.
3. Create a NumPy array containing the number of bedrooms, square footage, and sale price of each house.
4. Select houses with more than four bedrooms using the condition house_data[:, 0] > 4.
5. Extract the sale prices of the selected houses.
6. Calculate the average sale price using np.mean().
7. Display the average sale price.
8. Stop the program.

Code

```
import numpy as np

# House data: Bedrooms, Square Footage, Sale Price

house_data = np.array([

   [3, 1500, 200000],

   [5, 2500, 350000],

   [4, 1800, 250000],

   [6, 3000, 450000],

   [5, 2200, 400000]

])

# Select houses with more than 4 bedrooms

selected_houses = house_data[house_data[:, 0] > 4]

# Extract sale prices and calculate their average

sale_prices = selected_houses[:, 2]
```

```
average_price = np.mean(sale_prices)

print("Average sale price of houses with more than 4 bedrooms:",
    average_price)
```

Input

Bedrooms	Square Footage	Sale Price
3	1500	200000
5	2500	350000
4	1800	250000
6	3000	450000
5	2200	400000

Output

Average sale price of houses with more than 4 bedrooms: 400000.0

```
========= RESTART: C:\Users\sivak\Desktop\FODS LAB experiments\exp 3.py ========
Average sale price of houses with more than 4 bedrooms: 400000.0
```

Result

Thus, houses with more than four bedrooms were selected using NumPy conditional filtering, and their average sale price was calculated as 400000.00.